Production probability, transport, and grammatical possibility

Reanalysing the English dative alternation with open data

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Abstract

This draft develops a no-new-data reassessment of Bresnan et al. (2007). The current analyses reproduce the production-choice target from `languageR::dative`, add explicit null and partial-pooling checks, and test whether the resulting conditioning profile transports to the Spoken BNC2014 dative dataset. The working result is that modern modelling preserves the classic production conclusion, while BNC2014 provides positive but partial cross-corpus transport evidence. The manuscript uses that result to keep apart production probability, acceptability preference, and grammatical possibility.

Keywords: dative alternation; probabilistic grammar; grammaticality; corpus linguistics; acceptability

1 The problem

The English dative alternation is a small problem with a large methodological payoff. Speakers can say "give Kim the book" or "give the book to Kim", but the choice is uneven. It varies with verb, length, animacy, definiteness, pronominality, discourse status, corpus, and register.

That unevenness is easy to misread. A rare pattern can be lexically restricted, strongly dispreferred, or tied to a discourse context without falling outside the grammar. A production model therefore has two jobs. It has to predict attested choices, and it has to say what kind of grammatical claim those choices support.

Bresnan et al. (2007) made the case that corpus probabilities can reveal grammatical possibilities that informal judgements miss. This paper asks how that case looks after open-data replication, modern validation, and cross-corpus transport.

The question is one of PROJECTIBILITY: what observing a conditioning profile in one setting licenses us to predict about another (Goodman, 1955). For the dative alternation, the relevant settings are a packaged reanalysis dataset, a later spoken British corpus, and possible acceptability evidence. Keeping those settings apart is the paper's main discipline.

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2 The original target

Bresnan et al. (2007) is the anchor because it treats the alternation as a grammatical problem rather than as a simple frequency count. The claim is not that whatever occurs often is grammatical. The claim is that observed choices can expose systematic constraints on what speakers treat as available.

That move matters because informal judgements often compress several facts into one verdict. A construction may sound odd because it's impossible, because it is rare, because the wrong verb is present, because a discourse condition is missing, or because the constructed example strips away the context that would license it. Corpus data can help separate these cases.

The empirical target is production choice. In Bresnan et al.'s setup, the response variable is whether the recipient appears as an NP in the double-object pattern or as a PP in the prepositional dative. The predictors describe properties of the verb, recipient, theme, discourse status, and corpus source.

This paper preserves that target. It treats production probability as evidence to be interpreted, rather than as a direct measure of grammaticality. The empirical question is whether the production profile survives cleaner modelling and whether it travels beyond the original corpus setting.

3 Data and direct replication

The analysis starts with `languageR::dative`, distributed with the CRAN `languageR` package (Baayen & Shafaei-Bajestan, 2025). The analysis scripts fetch the package source tarball, load the packaged `dative.rda` object, and write derived summaries.

In the packaged dataset, there are 3,263 rows and 15 columns. The response records recipient realization, with 2,414 NP realizations and 849 PP realizations. The dataset contains 75 verb levels and two modalities: 2,360 spoken tokens and 903 written tokens. Speaker identifiers are missing exactly for the written rows.

For the first model sequence, the response stays binary: NP is coded as 1 and PP as 0. The initial split is verb-stratified, with a fixed seed (`20260623`), so every verb observed in the test set is also represented in training. That split gives 2,637 training rows and 626 test rows.

The denominator is therefore narrow. These rows are already tokens in which a dative recipient was realized as NP or PP. They aren't all contexts in which a transfer event could have been expressed, omitted, paraphrased, or packaged in some other construction. The models estimate conditional choice within an attested alternation frame: given that a token enters the dative-alternation sample, which realization appears?

Initial analyses are deliberately plain. A marginal model sets the baseline. Structured models then add modality, length, verb, and the non-verb predictors. Two falsification checks accompany them: a scrambled-label check and fake-data checks. These tests ask whether the fitted structure beats chance, not whether it proves a grammatical theory.

A second empirical source is the Spoken BNC2014 dative dataset (G. Jenset & McGillivray, 2018; G. B. Jenset & McGillivray, 2019). The verified Figshare CSV has 1,839 parsed data rows and 44 columns under a CC BY 4.0 licence. The apparent line-count discrepancy is mechanical: the file has 1,839 newline characters but no terminal newline, giving 1,840 logical text lines including the header.

BNC2014 serves as a transport target rather than a direct replication of `languageR::dative`. Its outcome field is `Pattern`, where `VNN` maps to NP realization and `VNPP` maps to PP realization. The six shared verbs are *give*, *send*, *show*, *sell*, *offer*, and *lend*. All six occur in the packaged `languageR` dataset.

4 Modern modelling update

The `languageR` result is a robustness result. The central question is whether modern modelling changes Bresnan et al.'s linguistic conclusion or mainly repackages it statistically. The answer, so far, is the latter.

Model	Test log loss	Test AUC	df
Marginal null	0.554	0.500	1
Non-verb main model	0.271	0.932	18
Fixed-verb full model	0.234	0.951	92
Hierarchical verb-intercept model	0.233	0.952	19

Table 1: Held-out `languageR::dative` results for the main model sequence.

The non-verb model already predicts held-out production choices well. Adding fixed verb effects improves test log loss from 0.271 to 0.234 and raises AUC from 0.932 to 0.951. The improvement confirms that lexical conditioning is not a minor residue after length, animacy, definiteness, pronominality, and accessibility are included.

The hierarchical verb-intercept model reaches almost the same held-out performance with far fewer effective parameters. Its test log loss is 0.233 and its AUC is 0.952. The estimated verb random-intercept standard deviation is 2.106, so the lexical signal remains large under partial pooling.

The null checks behave as they should. Scrambled-label fits lose predictive value. Fake-data fits without a verb signal collapse toward singular zero-variance random-effects models, while a fake verb-effect check recovers a nonzero verb standard deviation. These checks make the production result more credible, but they do not change its explanatory level.

The result supports Bresnan et al.'s production-choice conclusion. It does not create a new grammatical conclusion on its own. The same-data analysis shows that the classic profile is reproducible and statistically stable. The paper's new empirical pressure has to come from transport.

5 Generalization and acceptability

The BNC2014 analysis asks whether the production profile travels across corpus design, variety, period, and annotation regime. The transport model is trained on the spoken `languageR` rows restricted to the six BNC verbs. It's then evaluated on harmonized BNC2014 rows.

Harmonization is lossy. The outcome maps cleanly: `VNN` becomes NP and `VNPP` becomes PP. Verb also maps cleanly after restricting the `languageR` data to the six shared verbs. Length is more delicate: BNC2014 releases character lengths, while `languageR` uses token-like lengths, so the transport script derives BNC word-count proxies in memory.

Several predictors are available only as proxies. BNC2014 has recipient and theme animacy, theme definiteness, and recipient/theme pronominality, but some fields have missing values and different annotation paths. Recipient definiteness is not released as a field, so it enters only as a labelled regex proxy in a sensitivity check. Discourse accessibility is unavailable and is dropped from transport.

The transported core model beats the marginal baseline by a wide margin. Its test log loss is 0.308, compared with 0.473 for the `languageR` marginal baseline, and its AUC is 0.906. When the BNC outcome is scrambled, the same predictions collapse to 0.885 log loss and 0.515 AUC.

Model	Test data	Log loss	AUC
languageR marginal	BNC2014	0.473	0.500
languageR core transport	BNC2014	0.308	0.906
languageR core transport	Scrambled BNC2014	0.885	0.515
BNC2014 marginal holdout	BNC2014	0.481	0.500
BNC2014 native core holdout	BNC2014	0.202	0.960
BNC2014 native scrambled train	BNC2014	0.495	0.440

Table 2: First BNC2014 transport and native holdout comparisons.

The BNC-native model is still stronger. On a verb-stratified BNC holdout split, the native core model reaches 0.202 log loss and 0.960 AUC. This gap matters. The transported model has learned a profile that's genuinely useful in the new corpus, but the BNC data also contains local structure that the `languageR` fit does not capture.

Several directions travel. Recipient length is negative for NP realization; theme length is positive; recipient pronominality is strongly positive; theme pronominality is strongly negative; `sell` and `send` are less NP-like than `give`. These are the strongest signs that the Bresnan profile is projectible beyond one packaged dataset.

Some details do not travel cleanly. The transported model underpredicts NP realization for `send`, `offer`, and `lend`, and overpredicts it for `sell`. Theme definiteness is the clearest coefficient mismatch: negative in the `languageR`-trained model, positive and weak in the BNC-native model.

The recipient-definiteness proxy should stay a sensitivity check. Adding it worsens transport log loss from 0.308 to 0.347 while leaving AUC essentially unchanged. That pattern fits the harmonization note: the proxy is interpretable, but it should not carry the headline result.

Acceptability evidence answers a different question. DAIS, introduced by Hawkins et al. (2020), contains human judgements for 5,000 dative sentence pairs and systematically varies verb, definiteness, and argument length. It's useful as a future bridge from production probability to acceptability preference. It's not analyzed here: the public repository has no explicit licence file, and the outcome is judgement preference rather than corpus choice.

6 Grammatical possibility

The empirical result supports a sharper vocabulary, not a shortcut from frequency to grammar. The `languageR` analysis shows that a classic production profile is reproducible under explicit validation. The BNC2014 analysis shows that much of that profile travels to a later spoken British corpus. Both results concern production choice.

Production choice is not the same as acceptability preference. A corpus token records what a speaker or writer did in a context. A judgement task asks how speakers compare alternatives under controlled conditions. These measures can align, but alignment is an empirical result rather than a definition.

The opportunity set matters. The present corpus evidence is conditioned on attested dative-alternation tokens: cases already annotated as NP or PP recipient realizations. It can support claims about which realization speakers chose inside that frame. It cannot by itself estimate how often speakers might have used a non-dative paraphrase, left the event implicit, or accepted the less frequent member of a constructed pair.

Grammatical possibility is a third target. It concerns whether a construction belongs to the grammar, including cases that are rare, strongly conditioned, or unattested in a finite corpus. Production data can support claims about that target only through an explicit bridge from attested choices to possible alternatives.

The dative alternation makes the bridge visible. A low-probability double-object realization may be grammatical but dispreferred under the observed predictors. A prepositional realization may be likely because the theme is short and the recipient is long. A form may be absent because the corpus lacks the relevant verb, register, or discourse setting.

The current analyses therefore license a cautious claim. Bresnan et al.'s production-conditioning profile is durable and partly transportable. That supports the idea that corpus probabilities can reveal structure that informal judgements miss. It does not make production probability identical to grammatical possibility.

7 Conclusion

The reanalysis has two empirical results. First, the `languageR` profile survives baselines, falsification checks, and partial pooling. The hierarchical model matches the fixed-verb model with a smaller effective parameterization, so modern modelling confirms the classic production-choice conclusion.

Second, the profile partly transports. A spoken six-verb model trained on `languageR` predicts BNC2014 well above a marginal baseline and fails against a scrambled BNC outcome. The BNC-native model still performs better, showing that transport is real but incomplete.

That combination is the paper's centre. The classic result is not overturned. It's made reproducible, tested against real nulls, and moved into a second corpus where it can succeed or fail. The success is strong enough to matter, and the failures are specific enough to guide the next analysis.

The conceptual lesson is the same one that motivates the title. Production probability is evidence about grammatical possibility, not a synonym for it. Open data and transport tests make that evidence stronger, but they also mark the boundary of what production models can show without acceptability data.

Data and code availability

The analysis scripts fetch public datasets to temporary locations. Derived files live in `data/derived/`; source and licence notes live in `notes/source-verification.md`. DAIS is cited as a possible acceptability bridge, but its files are not fetched, analyzed, or redistributed here because the public repository does not include an explicit licence file.

Acknowledgements

TODO: Add final AI-use disclosure with the actual model list used on the project.

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